

ELI — Learning

ELI v2.5.9

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ELI Learning — LoRA Self-Training Pipeline

Updated for v2.4.84. Stage 12 learning hooks for chat turns are centralized in `eli/cognition/learning_coordinator.py (finalize_turn())`.

`eli/learning/` — 4.3k LOC, 14 files. ELI's self-improvement-via-fine-tuning path. **Important framing:** this is a *curated, human-gated, operator-invoked* training pipeline — **not** an autonomous self-modifying loop, and it targets a **separate trainable Hugging Face base**, not the inference GGUF.

It is driven from **Labs ► Training** (a four-step wizard) as well as from chat actions and the overnight scheduled task. Two long-standing blockers were removed at the same time: the human review gate now has an interface, and the target system is no longer locked to Phi-3.

Files & the pipeline

The stages, roughly in order:

File	LOC	Stage
<code>dataset_builder.py</code>	552	turn ELI's logged turns/corrections into supervised examples
<code>dataset_filters.py</code>	148	quality gates (<code>is_bad_response</code> , <code>row_is_reviewed</code>)
<code>export_trainable_dataset.py</code>	162	export reviewed rows → trainable JSONL
<code>merge_reviewed_datasets.py</code>	129	merge reviewed dataset shards
<code>bootstrap_phi3_base.py</code>	239	one-time download of the trainable HF Phi-3 base
<code>base_model_resolver.py</code>	251	locate/validate the trainable base (vs an adapter)
<code>training_preflight.py</code>	136	check peft/transformers/datasets present + target ready

File	LOC	Stage
<code>lora_trainer_guard.py</code>	630	TrainerTarget plans (paths, base_family) + guard checks
<code>lora_trainer.py</code>	834	device selection, QLoRA, chat-template prompts, PEFT loop
<code>lora_eval.py</code>	485	eval harness (score vs expected/forbidden, inspect adapter)
<code>target_registry.py</code>	266	operator-declared training targets, any model family
<code>review_queue.py</code>	270	the human review gate: triage, approve/edit, write trainable rows

Flow

1. **Build** (`dataset_builder.py`): mines ELI’s own conversation/correction history into SupervisedExamples. Crucially includes `clean_text` + **redact_text (PII redaction)** and an exclusion set so sensitive/garbage content never becomes a training example. Writes `training/datasets/eli_supervised_v0.jsonl` + a report.
2. **Gate** (`dataset_filters.py`): `is_bad_response` and `row_is_reviewed` enforce quality + a **human review flag**.
3. **Export / merge**: reviewed rows → `*.trainable.jsonl`.
4. **Preflight** (`training_preflight.py`): refuses to proceed unless the HF training stack is installed and the target/base resolve.
5. **Train** (`lora_trainer.py`): `_dataset_report` **refuses to train** on a dataset with unreviewed rows, wrong-target rows, or bad-response rows — i.e. it will not learn from un-vetted data. Loads the base through native transformers (with a RoPE-scaling compat shim for Phi-3), trains the adapter. Two things changed in 2.3.7:
 - **Prompt format follows the base model.** `_format_example` used a hardcoded Phi-3 `<|user|>/<|assistant|>` literal. Training a Qwen or Llama checkpoint on Phi-3 turn markers teaches it tokens its chat template never uses, degrading the model instead of tuning it. The tokenizer’s own chat template is now the source of truth, with the literal kept as a fallback for bases without one.
 - **QLoRA.** With bitsandbytes present the frozen base loads in 4-bit (nf4, double quant, fp16 compute) with a paged optimiser, so a card too small to hold the fp16 weights can still train the adapter.
6. **Eval** (`lora_eval.py`): `score_response` against expected/forbidden, plus `inspect_adapter` / `inspect_eval_suite`.

Device selection (`lora_trainer._pick_device`)

The old rule was a flat `free_vram >= 10 GiB` floor written for a Phi-3 on an 8 GB card, and its failure mode was a **silent** fall-through to CPU — where a run takes days and looks exactly like a hang. It also named every accelerator “cuda”, showing CUDA on a Radeon.

- `_accelerator()` reports the vendor honestly. torch routes ROCm through the `torch.cuda` API (a HIP build answers `torch.cuda.is_available()`), so one code path serves NVIDIA and AMD; the vendor is read from `torch.version.hip`. Apple (mps) and Intel (xpu) are detected and **honestly marked unable** to run this trainer rather than being silently ignored.
- `estimate_vram_gb()` sizes the requirement from the checkpoint’s real size on disk, so the floor holds for a 1B and for a 35B instead of a hardcoded figure.
- A refusal states what would fix it. On a 2060 Super: *“6.37 GiB free but this run needs ~9.04 GiB ... free VRAM, shorten the sequence length, or install bitsandbytes for 4-bit training.”*

Paths

`eli.core.paths.learning_dir()` follows the `models_dir()` pattern: the source tree in dev so the existing training/ layout keeps working, the **user data dir** on a packaged install where `project_root()` is a read-only mount. Run logs, guard plans and the target registry all follow it, so a frozen build no longer writes into its own installation.

Targets (`target_registry.py` + `lora_trainer_guard.py`)

A **target** pairs a base model with its dataset, adapter and output directory. Until 2.3.7 the allowlist was the literal set `{eli_phi, eli_phi_ultra}` and the guard asserted `base_family == "phi3"`, so a redistributed copy running Qwen or Llama was refused at the first gate with no way to declare a target short of editing source.

The allowlist is still an allowlist — nothing trains unless it was explicitly declared — but declaring is now a **runtime act**:

- `target_registry.create_target(name, base_model_path, ...)` writes an operator target into `<data dir>/training/targets.json`. Built-ins remain, so existing installs are unaffected.
- **Family is read, not asserted.** `detect_family()` reads `model_type` from the base model's own `config.json`. The guard then checks the declared family against what is actually on disk, so a target claiming qwen3 over a Llama checkpoint is still refused. Nothing in the registry enumerates known families, so a model type released after this build still resolves.
- `base_model_resolver.discover_base_models()` scans the model roots for **any** trainable HF directory and reports each with its family and size.
- A target that has never trained has no adapter yet. That is now a *first run*, not a fault — only a malformed adapter is an error.

The review gate (`review_queue.py`)

`lora_trainer_guard` has always required every training row to be approved by a person and scoped to the target. That contract was enforced but **unreachable**: `dataset_builder` writes candidates tagged `needs_review`, and nothing in the product could retag one. Live state before this module existed: 615 candidate rows, 0 trainable, `will_train` permanently false.

`ReviewQueue` is the queue behind the Training tab:

- **Assisted triage** pre-rejects the provably unusable — `is_bad_response` matches, duplicates, instructions under 8 characters — and *flags* rows that need a human eye (very long output, redacted paths, heavy structure).
- **Triage never approves.** Approval is a person's act; there is a test pinning this. "Approve all clean rows" is one click standing for reading a filtered list, not an auto-approve.
- Editing a row **re-triages** it: an edit can rescue a rejected row and can equally break one that passed.
- `save()` writes approved rows tagged reviewed and scoped to the target — exactly the shape `_validate_dataset_rows` demands — overwriting rather than appending, so a second pass does not duplicate.

On the live corpus: 615 candidates → 20 auto-rejected, 309 clean, 286 flagged for judgement; approving the clean set took `train_ready` from false to **true** for the first time.

What it is — and isn't

- It **is**: a responsible, reproducible fine-tuning pipeline with PII redaction, human review gating, preflight, and an eval harness. The dataset comes from ELI's real interactions (corrections, failures, conversations).
- It **isn't**: (a) automatic — it is driven by the operator, through the Training tab, the chat actions, or a scheduled overnight job. (b) connected to the live inference model by itself — it trains an

adapter on a Hugging Face base, while inference runs a GGUF. A trained improvement reaches the running assistant only once the adapter is merged into the base and converted to GGUF (`training/merge_and_convert.py`); the Training tab says so explicitly when a run finishes rather than implying the model has changed.

Honest assessment

- **Strong (and unusually responsible):** PII redaction, refusing to train on unreviewed/bad rows, a preflight, and an eval suite are exactly what a serious fine-tuning loop needs and what most hobby projects skip. The base-vs-adapter validation prevents the classic “trained on an adapter” mistake.
- **Weak / watch:**
 1. **Base/inference disconnect** — training improves Phi-3, but the assistant usually runs another GGUF, so the self-improvement doesn’t reach the live model. The loop is only closed if you actually serve the trained model.
 2. **Manual** — “self-training” is operator-driven, not autonomous. That’s safe, but it means it improves only when you run it. (Arguably the right trade-off, but worth stating plainly rather than implying autonomy.)
 3. **Heavy deps** — needs the full HF/peft/transformers stack. `peft` and `datasets` were previously only in `requirements.lock.txt`, so a plain `pip install -r requirements.txt` produced an install whose Training tab could never leave preflight; both are now in `requirements.txt`, with `bitsandbytes` added for 4-bit training (excluded on macOS, which has no wheel).
 4. **Phi-3 remains the shipped default** — the built-in targets and `bootstrap_phi3_base.py` are still Phi-3 — but it is no longer a lock. Any local HF base can be declared as a target from the Training tab, and the prompt format follows the base model’s own chat template rather than Phi-3’s literal turn markers.
 5. **Merge-to-GGUF is still manual.** `training/merge_and_convert.py` closes the loop but depends on `unsloth`, which is in no requirements file. Producing a usable model after a successful run is therefore still a documented manual step, not a button.